XMachines' autonomous robot X100 at work in a farm (Courtesy: XMachines)

used Cropin's solution to digitise their hazelnut farms in Italy, while Unilever used it to digitise coconut farms in Indonesia and support the farmers with agro advisories and best practices. A large Indian business conglomerate, which deals with spice farms spread over 35,000 acres, and managed by 10,000 farmers across 170 villages, used Cropin's solution to digitise farmer and farm data, and to geotag it.

Similarly, one of the largest tea corporations based out of India, which produces 10 million kilograms of tea annually, partnered with CropIn to digitise their farm operations covering 50+ plantations and 2500+ acres of area under cultivation. For the Rainforest Alliance, Cropin developed CocoaSense, an AI-powered solution that leverages satellite imagery to efficiently manage cocoa crops in Ghana.

The Sustainable Livelihoods and Adaptation to Climate Change (SLACC) agency of the World Bank partnered with the government of India to improve the livelihoods of smallholder farmers. Cropin was chosen as a technology partner for this project, which aimed to provide climate-resilient agriculture to farmers in four flood and drought-affected districts in Madhya Pradesh and Bihar. The SLACC-Cropin project has positively impacted more than 8,000 farmers in the districts it was adopted in.

DeHaat is another full-stack agriculture platform provider, whose solutions span across the supply chain, from sourcing of farm requirements and soil testing to tech based advisory and selling of produce.

Unnati Agri, a fintech-driven agriculture service platform, raised US$8 million in a Series A round during FY22. It provides financial management services to farmers to solve their working capital requirements and supports them throughout their farming cycle. Registered farmers can buy seeds, pesticides, fertilisers and other products from sellers listed on Unnati. On the other hand, they can also sell their produce directly to food processors and agri-businesses on the same platform.

Farm-to-fork. Aibono specialises in aggregation and just-in-time supply of fresh fruits and vegetables. They specialise in farm-to-fork or 'kisan-to-kitchen' supply of agricultural produce directly from farms to consumers, with complete traceability.

Ninjacart also focuses on the supply chain. They connect producers of food directly with retailers, restaurants, and service providers using in-house applications that drive end-to-end operations. They currently operate in seven cities, and their supply chain is equipped to move 1,400 tonnes of perishables from farms to businesses, every day, in less than 12 hours.

Ecozen is another farm-to-fork venture founded by an IIT Kharagpur alumni, which aims to optimise the way perishables are handled across the value chain, with clean and innovative technology. ReshaMandi is also an end-to-end platform, but focuses on the natural fibres industry—it is farm-to-fashion, rather than farm-to-fork!

AgroStar, Otipy, Poshn, Arya.ag, Apna Godam, Agrowave, Bijak, Crofarm, Clover Ventures, Ergos, Farm2Fam, and WayCool are other food supply chain startups to watch.

Traceability. TraceX builds blockchain powered traceability solutions that can be used to build transparency and trust into the agricultural supply chain. It can be used for a variety of purposes, from seed traceability to managing contract farmers.

Their TraceCarbon is a blockchain-powered carbon management platform, which helps companies to decarbonise Scope 3 emissions, in their journey towards being carbon neutral. Scope 3 emissions are those emissions that are not actually produced by the directly-owned assets or activities of the company, but those produced up or down the value chain, but for which the company is responsible. For example, emissions caused by contract farmers.

Solutions from the younger generation

It is interesting to see students developing solutions to solve problems faced by farmers, often their own kith and kin, in remote rural areas. This kind of connect to their roots gives us hope that the next generation will help revive the agricultural sector using the right technologies.

Kishan Know is one such indigenous AI solution developed by three young students at the Vellore